

Documentation Staleness Audit

Read-only audit — no doc changes made yet. Cross-references git commit history against current source to flag stale/superseded documentation.

Key event that dates the docs: a **Katapult bootloader** was ported to the STM32G474 and made STM32 wireless/SD flashing “rock solid.” Several docs were written before Katapult and describe the old (failed) ROM-bootloader reality.

The timeline that matters (git commit dates)

Date	Commit	Event
6/15	<code>0c0c279</code>	KNOWN_ISSUES.md written — “SD flash impossible, use ST-LINK”
6/17	<code>c79db17</code>	SD_AND_STM32_FLASHING_FEASIBILITY.md — “analysis only, nothing implemented”
6/20	<code>b65d428</code>	WIRELESS_STM32_FLASH_EXPLAINED.md — ROM/BOOT0 software-entry saga
6/20	<code>b056943</code> +	Katapult Phase 1 & 2 ported + implemented
6/21	<code>29be8a0</code>	“Fix all Katapult protocol bugs causing STM32 bricking” → Katapult working
6/21	<code>48e3164</code>	UART moved to UART3-IN header (GPIO 27/28) @ 576 kbaud
6/22	<code>3311f60</code>	SD-card firmware flashing added for ESP32-P4 + STM32

Proven current reality (from source, not docs):

- App relocated: `STM32G474CETX_FLASH.ld` → `FLASH_ORIGIN = 0x8002000` (8 KiB Katapult reserve)
- `main.c`: `SCB->VTOR = 0x08002000U`; + banner `BOOT,FW=KATAPULT-APP-v6-RELOCATED-0x08002000`
- ESP32-P4 link: `STM_APP_BAUD 576000` on GPIO 27/28 (UART3-IN); `KATAPULT_BAUD 250000`
- Katapult binaries committed in `STM32G474CE/katapult/`



Verdicts

Doc	Verdict	Why
docs/issues/KNOWN_ISSUES.md	 STALE — rewrite or archive	Written 6/15, 6 days before Katapult. Whole premise (SD flash impossible, no BOOT0, ROM won't work via software, ST-LINK required, option-byte unverified) was solved by Katapult.
docs/features/SD_AND_STM32_FLASHING_FEASIBILITY.md	 ARCHIVE (historical)	Pure feasibility study that says “nothing implemented.” It has since been implemented via Katapult. Still lists old GPIO17/18 wiring. Valuable as history, not as current guidance.
docs/features/WIRELESS_STM32_FLASH_EXPLAINED.md	 ARCHIVE (historical)	Documents the abandoned ROM-bootloader / AN-3155 / BOOT0+NRST approach (GPIO29-32, 1 Mbaud, ROM @ 0x1FFF0000, 8E1 parity, breadcrumb). Superseded wholesale by Katapult. Great post-mortem, but not how it works now.
docs/firmware/KATAPULT_BOOTLOADER_SETUP.md	 CURRENT — minor update	This is the authoritative, correct doc. Two stale details: says app link is “1,000,000 baud” (now 576000) and Katapult validate step shows <code>-b 250000 on /dev/ttyUSB0</code> which is fine, but the app-baud/GPIO (27/28) should be noted.
README.md	 UPDATE — stale sections	Lines ~61-110: “STM32 update via ST-LINK,” “SD-Card flashing requires ST-LINK,” “experimental option-byte... unverified.” All pre-Katapult. Should now point to Katapult wireless/SD flashing.
docs/hardware/PIN_ASSIGNMENTS.md	 CURRENT	STM32-side pinout (PA9/PA10, NRST) unchanged by Kata-

Doc	Verdict	Why
		pult. The GPIO move was on the ESP side.
docs/hardware/BALANCER_SPECIFICATION.md	● CURRENT	Hardware spec, no flash dependency.
docs/hardware/SNUBER_SPECIFICATION.md	● CURRENT	Hardware spec.
docs/hardware/BUCK_CONVERTER_PIN_ANALYSIS.md	● CURRENT	Hardware analysis.
docs/hardware/FIRMWARE_MEASUREMENT_ANALYSIS.md	● CURRENT	Measurement analysis.
docs/firmware/ESP32_ARCHITECTURE_ANALYSIS.md	● CURRENT	Architecture; not flash-specific.
docs/firmware/JOULE_ALGORITHM_FIX.md	● CURRENT	Algorithm history.
docs/firmware/MODE_SWITCHER_AND_JOULE_CLEANUP.md	● CURRENT	Feature history.
docs/firmware/STATUS_TAB_UI_IMPROVEMENTS.md	● CURRENT	UI history.
docs/firmware/CHANGELOG_P4_UI_STABLE.md	● CURRENT	Changelog.

Proposed actions (for approval — nothing done yet)

1. **Create docs/archive/** and `git mv` the two superseded flash docs there:
 - docs/features/WIRELESS_STM32_FLASH_EXPLAINED.md → docs/archive/
 - docs/features/SD_AND_STM32_FLASHING_FEASIBILITY.md → docs/archive/
 - Add a one-line “SUPERSEDED BY KATAPULT_BOOTLOADER_SETUP.md” banner at top of each.
2. **KNOWN_ISSUES.md** — either rewrite the STM32-flash section to reflect Katapult (recommended) or archive it with a superseded note. Your call on which.

3. **KATAPULT_BOOTLOADER_SETUP.md** — fix the app baud (1,000,000 → 576000) and note the ESP link is GPIO 27/28 (UART3-IN).
4. **README.md** — replace the “ST-LINK required / SD unreliable / option-byte unverified” sections with the current Katapult wireless + SD flow.
5. Add a note in `docs/archive/README.md` explaining these are kept for history.